

ARUL RHIK MAZUMDER

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Education

Carnegie Mellon University

Aug. 2023 – May 2026

Bachelor of Science in Computer Science, Dean's List High Honors

Pittsburgh, Pennsylvania

Select Coursework: Probabilistic Graphical Models, Quantum Computing, Complexity Theory,
Parallel Architecture and Algorithm Design, Deep Learning Systems

Activities: President of Quantum Computing @ CMU, Quant Club, ACM @ CMU, Project Olympus

Experience

Caltech SURF

June 2025 - August 2025

Undergraduate Research Fellow

Pasadena, California

- Worked under Samson Wang with John Preskill's research group to design early fault-tolerant quantum algorithms based on Trotterization, achieving provable runtime guarantees.
- Engaged in a biweekly quantum algorithms reading group, exploring cutting-edge techniques such as Quantum Signal Processing (QSP) and Quantum Singular Value Transformation (QSVT).

Duke Software-Tailored Architectures for Quantum Codesign (STAQ)

June 2025

Invited Scholar

Durham, North Carolina

- Completed a PhD-level, week-long quantum computing seminar featuring advanced lectures on quantum simulation, error correction, and architecture, taught by leading researchers including Peter Love, Kenneth Brown, and Yu Tong
- Participated in a lab tour at Duke Quantum Center and learned about experimental research across diverse quantum hardware platforms, including ion traps and neutral atoms.

International Business Machines (IBM)

May 2024 – August 2024

Research Intern

Cambridge, Massachusetts

- Implemented a novel learning algorithm for discrete graphical models using samples generated from Glauber Dynamics.

Carnegie Mellon University Quantum Technologies Group

May 2023 – Current

Research Intern

Pittsburgh, Pennsylvania

- Produced tutorial paper that aids users in developing and implementing their solutions to their optimization problems on current IBM and D-Wave quantum machines. Work will be presented at INFORM Tutorials 2025.

Select Projects/Publications

Quantum Annealing for solving the Shipment Rerouting | *Python, D-Wave Ocean, MILP*

January 2025

- Developed both classical approximation algorithms and quantum annealing algorithms for SRP using D-Wave's quantum annealer and custom MILP models
- Demonstrated that quantum annealing solutions achieve near-optimal performance significantly faster than classical solvers in diverse logistics scenarios

Differential Evolution based Hyperparameters Selection of Transformer | *Python, Deep Learning*

August 2023

- Applied metaheuristics like Differential Evolution to select hyperparameters for a variety of deep learning models (ANN, LSTM, GRU, Transformers) to produce more accurate results when load forecasting
- Paper accepted by the 14th IEEE International Symposium Series on Computational Intelligence (SSCI 2023)

A Hybrid Quantum Algorithm for the Metric Traveling Salesman Problem | *Python, Ocean*

January 2023

- Researched and developed a novel quantum-hybrid algorithm for the Traveling Salesman Problem that performs three times faster on larger input sizes than the best classical and quantum algorithms. Tested and run on DWAVE backends
- Published in 2023 IEEE International Parallel and Distributed Processing Symposium (IPDPS)

Comparisons of Classic and Quantum String Matching Algorithms | *Python, Qiskit*

August 2022

- Developed novel quantum string-matching algorithm using Qiskit and tested it on IBM Quantum Backends
- Published in proceedings of ACM's 4th International Conference on Advanced Information Science and System (AISS)

Awards

Olympiads: USA(J)MO (2021-22), USAPhO Bronze (2022), USACO Gold (2023), USNCO National Finalist (2022)

Other Competitions: 2nd Place overall at YQuantum - The Yale Quantum Institute Grand Prize Recipient (2025),
2nd Place at Correlation One and Citadel Summer Invitational Datathon (2024),
Graduate Student Association Best Campus Infrastructure Hack Award at Hack@CMU (2023)
Regeneron Science Talent Search Scholar [Top 300] (2023), Purple Comet 5th Place (2022)
International Math Modelling Competition National Finalist (2022),
United States Math Talent Search Honorable Mention and Bronze (2021-22)